



杭州电子科技大学
HANGZHOU DIANZI UNIVERSITY



DFS

吴超

通信工程学院

复习

□ DTFT性质【时移、频移、卷积特性、Parseval定理】

□ 例 $x[n - n_0] \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$ $e^{j\omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X(e^{j(\omega - \omega_0)})$

□ DFS及IDFS定义

□ 例 $\tilde{x}[n] = \text{IDFS}\{\tilde{X}[k]\} = \frac{1}{N} \sum_{k=\langle N \rangle} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$ $\tilde{X}[k] = \text{DFS}\{\tilde{x}[n]\} = \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$

□ DFS性质【线性、时移、频移、对称特性、周期卷积】

□ 例如 $\text{DFS}\{\tilde{x}^*[n]\} = \tilde{X}^*[-k]$ $\text{DFS}\{\tilde{x}^*[-n]\} = \tilde{X}^*[k]$

□ 特殊的时域是实函数、实对称函数的频谱特点

周期卷积与线性卷积

$$\begin{aligned}\tilde{y}[n] &= \tilde{x}[n] * \tilde{x}[n] \\ &= \{1, 1, 0\} * \{1, 1, 0\} \\ &= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}y[n] &= x[n] * x[n] \\ &= \{1, 1, 0\} * \{1, 1, 0\} \\ &= (\delta(n) + \delta(n-1)) * (\delta(n) + \delta(n-1)) \\ &= \delta(n) + 2\delta(n-1) + \delta(n-2)\end{aligned}$$

周期序列DFS和IDFS定义

周期为 N 的任意序列 $\tilde{x}[n]$

$$\tilde{x}[n] = \text{IDFS}\{\tilde{X}[k]\} = \frac{1}{N} \sum_{k=\langle N \rangle} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$$

$$\tilde{X}[k] = \text{DFS}\{\tilde{x}[n]\} = \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

DFS的基本性质

3. 对称特性

$$\text{DFS}\{\tilde{x}^*[n]\} = \tilde{X}^*[-k]$$

$$\text{DFS}\{\tilde{x}^*[-n]\} = \tilde{X}^*[k]$$

- 若为实序列，则有

$$\tilde{X}[k] = \tilde{X}^*[-k]$$

$$\begin{aligned} |\tilde{X}[k]| &= |\tilde{X}[-k]| \\ \phi[k] &= -\phi[-k] \end{aligned}$$

$$\begin{aligned} \tilde{X}_R[k] &= \tilde{X}_R[-k] \\ \tilde{X}_I[k] &= -\tilde{X}_I[-k] \end{aligned}$$

$$\text{DFS}\{\tilde{x}^*[n]\} = \tilde{X}^*[-k]$$

5. 复序列的共轭

$$Z[x^*(n)] = X^*(z^*) \quad R_{x-} < |z| < R_{x+}$$

$$\tilde{X}[k] = \sum_{n=-\infty}^{+\infty} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

$$X[z] = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$\text{DFS}\{\tilde{x}^*[-n]\} = \tilde{X}^*[k]$$

6. 翻褶序列

$$Z[x(-n)] = X\left(\frac{1}{z}\right) \quad \frac{1}{R_{x+}} < |z| < \frac{1}{R_{x1}}$$

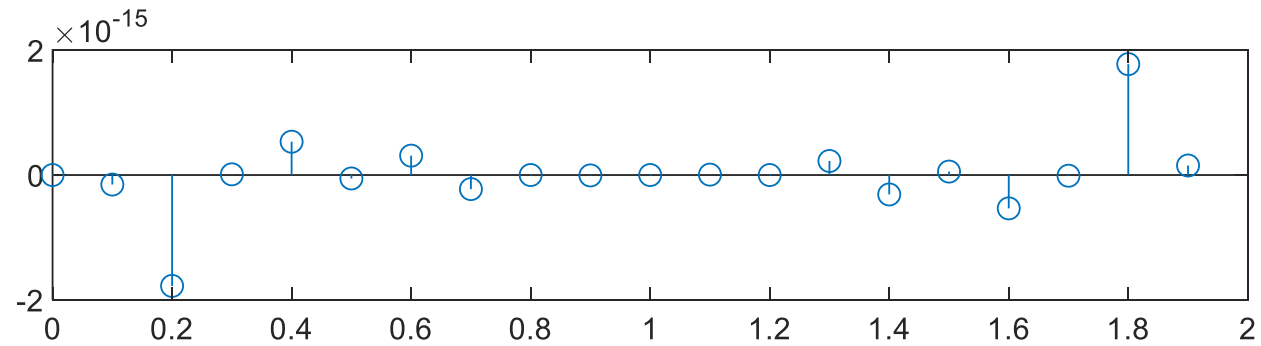
DFS的基本性质

- 若 $\tilde{x}[n]$ 为偶对称的实序列，则有

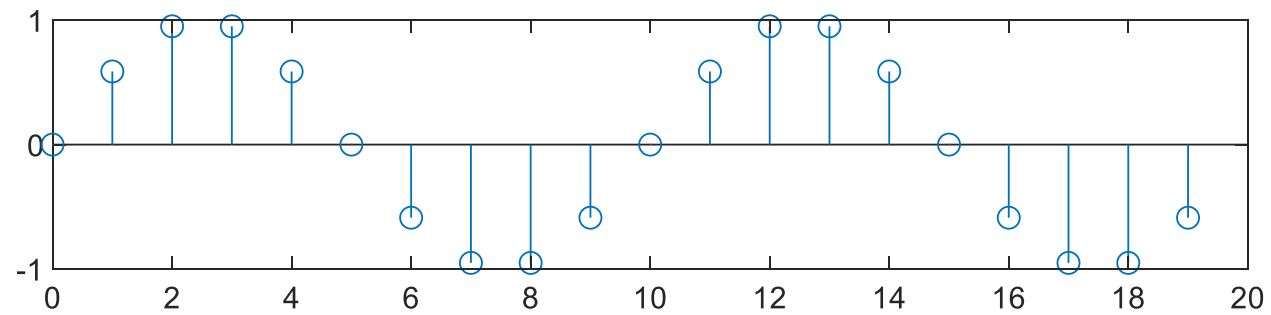
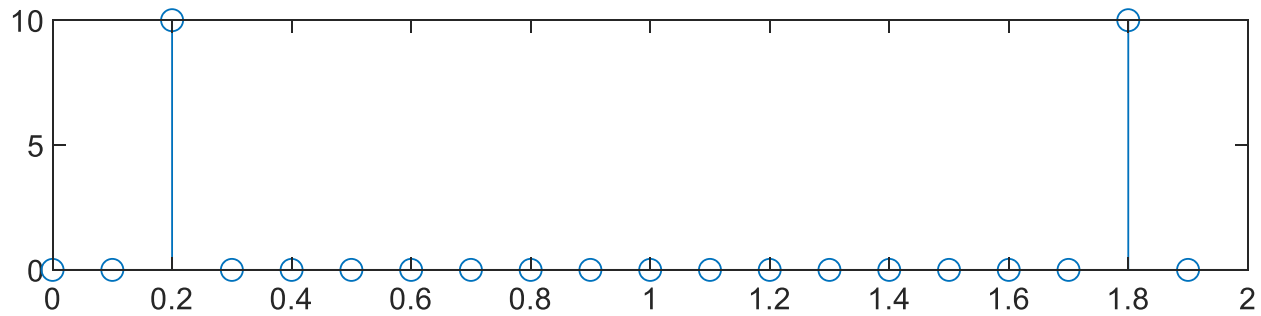
$\tilde{x}(k)$ 为实序列，且为偶对称

- 若 $\tilde{x}[n]$ 为奇对称的实序列，则有

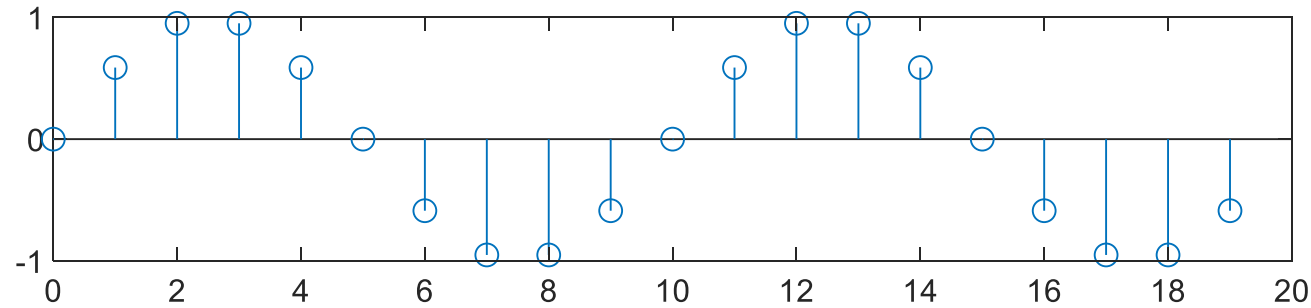
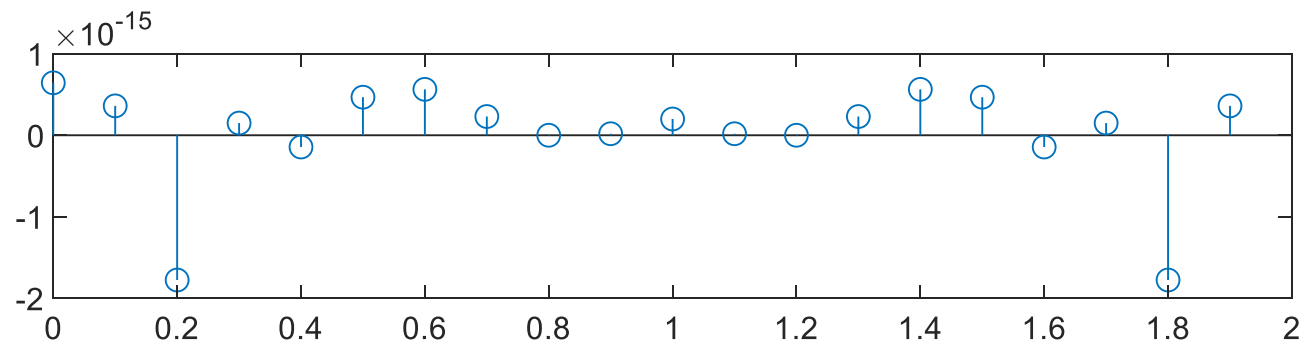
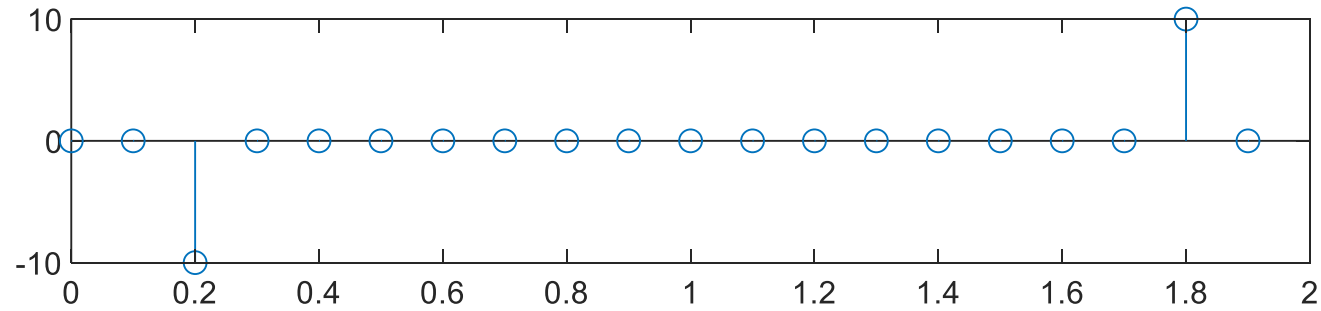
$\tilde{x}(k)$ 为纯虚序列，且为奇对称



$$\cos(\pi/5n)$$



$$\sin \left(\frac{\pi}{5n} \right)$$



DFS的基本性质

- 若 $\tilde{x}[n]$ 为偶对称的实序列，则有 $\tilde{x}(k)$ 为实序列，且为偶对称

$$\text{DFS}\{\tilde{x}^*[n]\} = \tilde{X}^*[-k] \quad \text{DFS}\{\tilde{x}^*[-n]\} = \tilde{X}^*[k]$$

$$\tilde{X}[k] = \tilde{X}^*[-k]$$

$$\tilde{X}[k] = \tilde{X}^*[k]$$

$$\tilde{X}_R[k] = \tilde{X}_R[-k]$$

$$\tilde{X}_I[k] = -\tilde{X}_I[-k]$$

$$\tilde{X}_I[k] = 0$$

- 若 $\tilde{x}[n]$ 为奇对称的实序列，则有 $\tilde{x}(k)$ 为纯虚序列，且为奇对称

$$\text{DFS}\{\tilde{x}^*[n]\} = \tilde{X}^*[-k] \quad \text{DFS}\{\tilde{x}^*[-n]\} = \tilde{X}^*[k]$$

$$\tilde{X}[k] = \tilde{X}^*[-k]$$

$$\tilde{X}[k] = -\tilde{X}^*[k]$$

$$\tilde{X}_R[k] = \tilde{X}_R[-k]$$

$$\tilde{X}_I[k] = -\tilde{X}_I[-k]$$

$$\tilde{X}_R[k] = 0$$

DFS的基本性质

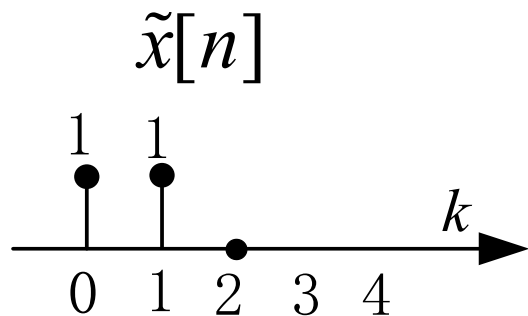
4.周期卷积定理

周期卷积定义：

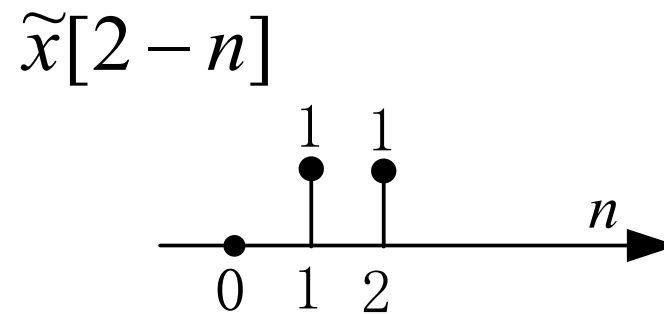
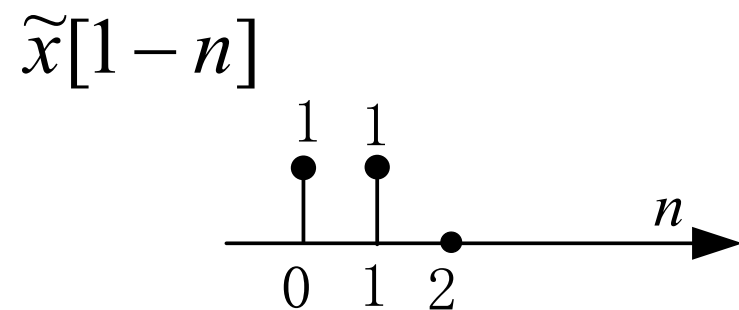
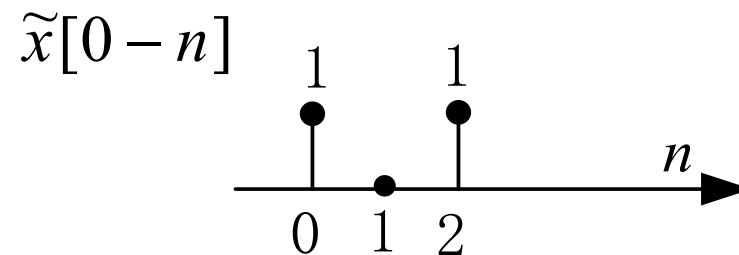
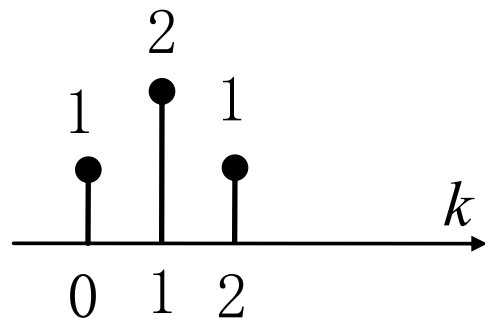
$$\tilde{x}_1[n] * \tilde{x}_2[n] = \sum_{n=0}^{N-1} \tilde{x}_1[n] \tilde{x}_2[m-n]$$

- 周期卷积是两个等周期的周期序列的卷积运算。
- 周期卷积的结果仍为相同周期的周期序列。

例：周期 $N=3$ 的序列 $\tilde{x}[n]$ 如图所示，试计算 $\tilde{y}[n] = \tilde{x}[n] * \tilde{x}[n]$



$\tilde{x}[n] * \tilde{x}[n]$



周期卷积的矩阵表示

$$\tilde{x}_1[n] * \tilde{x}_2[n] = \sum_{n=0}^{N-1} \tilde{x}_1[n] \tilde{x}_2[m-n]$$

例： $N=4$

$$\begin{bmatrix} \tilde{y}[0] \\ \tilde{y}[1] \\ \tilde{y}[2] \\ \tilde{y}[3] \end{bmatrix} = \begin{bmatrix} \tilde{x}_2[0] & \tilde{x}_2[-1] & \tilde{x}_2[-2] & \tilde{x}_2[-3] \\ \tilde{x}_2[1] & \tilde{x}_2[0] & \tilde{x}_2[-1] & \tilde{x}_2[-2] \\ \tilde{x}_2[2] & \tilde{x}_2[1] & \tilde{x}_2[0] & \tilde{x}_2[-1] \\ \tilde{x}_2[3] & \tilde{x}_2[2] & \tilde{x}_2[1] & \tilde{x}_2[0] \end{bmatrix} \begin{bmatrix} \tilde{x}_1[0] \\ \tilde{x}_1[1] \\ \tilde{x}_1[2] \\ \tilde{x}_1[3] \end{bmatrix}$$
$$= \begin{bmatrix} \tilde{x}_2[0] & \tilde{x}_2[3] & \tilde{x}_2[2] & \tilde{x}_2[1] \\ \tilde{x}_2[1] & \tilde{x}_2[0] & \tilde{x}_2[3] & \tilde{x}_2[2] \\ \tilde{x}_2[2] & \tilde{x}_2[1] & \tilde{x}_2[0] & \tilde{x}_2[3] \\ \tilde{x}_2[3] & \tilde{x}_2[2] & \tilde{x}_2[1] & \tilde{x}_2[0] \end{bmatrix} \begin{bmatrix} \tilde{x}_1[0] \\ \tilde{x}_1[1] \\ \tilde{x}_1[2] \\ \tilde{x}_1[3] \end{bmatrix}$$